6 In a cathode ray oscilloscope, an electron beam is fired from rest, with an accelerating voltage of 1.0 kV and strikes a fluorescent screen that is 14.0 cm from the end of the vertical deflection plates as shown in Fig. 6.1. Depending on the potential difference between the horizontal and vertical deflection plates, the beam can be manipulated to draw waveforms on the screen.

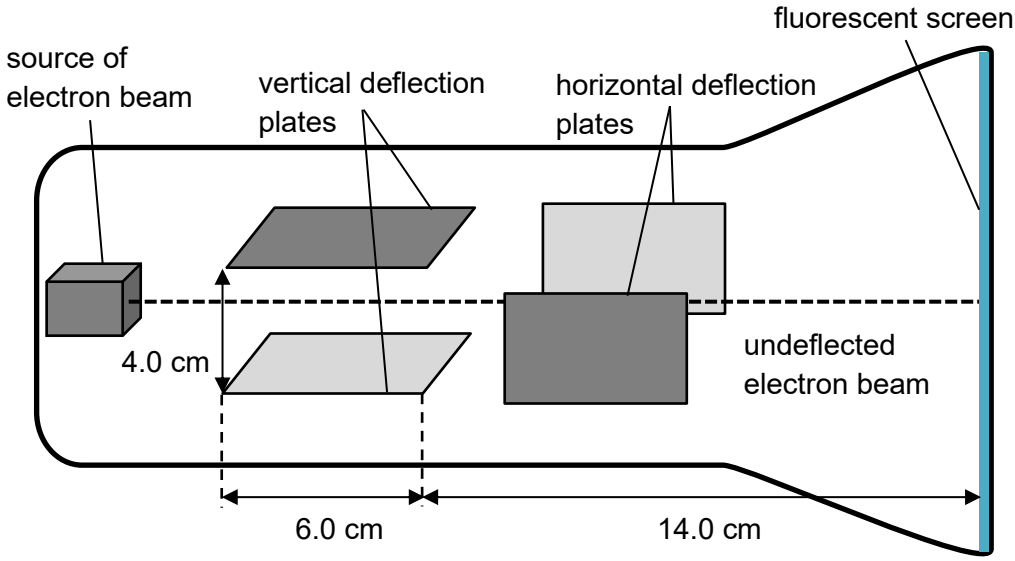

Fig. 6.1 (not to scale)

An alternating voltage is applied across the vertical deflection plates. The time-base is set at 5.0 ms per division and the vertical scale of the display is set at 5.0 V per division. The waveform on the screen is shown in Fig. 6.2.

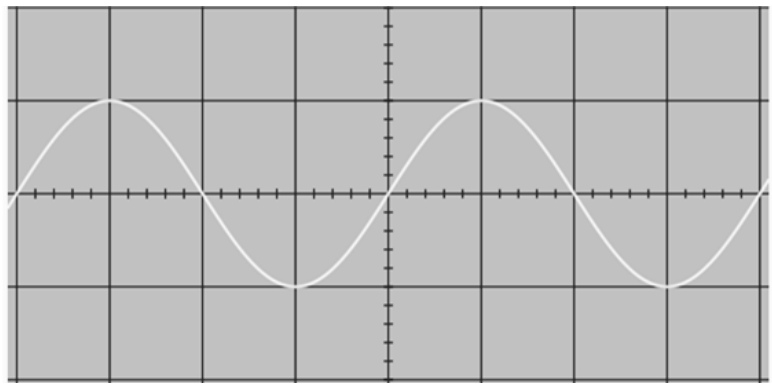

Fig. 6.2 (not to scale)

(a) Calculate the speed of the electrons just before they enter the region between the vertical deflection plates.

speed = …… m s^{-1} [2]

(b) Show that the time taken for an electron in the beam to travel between the vertical deflection plates is 3.2 ns.

[1]

(c) The actual applied voltage to the vertical deflection plates is 20 times that of the reading on the screen.

Determine the maximum vertical deflection of the electron beam when it reaches the fluorescent screen.

maximum vertical deflection = …… m [4]

(d) Describe and explain how the waveform in Fig. 6.2 would be different if the electron beam was replaced with a proton beam.

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……..……..

……..……..

……..…….. [2]